

Composition and Temporal Dynamics of Microbial Communities in Drinking Water from Peri-Urban Areas of Vellore.

Karthikeyan Govindan¹, Blossom Benny¹, Priyadharshini Karunakaran¹, Venkata Raghava Mohan², Dilip Abraham¹

¹ Wellcome Trust Research Laboratory, Division of Gastrointestinal Sciences, Christian Medical College, Vellore, Tamil Nadu, 632004

² Department of Community Health and Development, Christian Medical College, Vellore, Tamil Nadu, 632002



Background

- The presence of microorganisms in drinking water systems poses a potential threat, as some of these microbes may exhibit pathogenicity, leading to waterborne diseases. Traditional culture methods are limited, detecting only cultivable microorganisms, leaving much of the microbial community unexplored.
- In recent years, molecular techniques have emerged as powerful tools for studying microbial communities in diverse environments. Rapid detection and characterization of microbial communities through techniques like targeted and shotgun Metagenomics facilitates large-scale monitoring and surveillance of water supplies.

Aim

To study the dynamics and interspecies interactions shaping the microbial community structure in peri-urban drinking water using targeted Metagenomics on Illumina Miseq platform.

Methods

- Drinking water samples were collected from October 2021 to February 2022 as part of an environmental surveillance study from households at two different time points one month apart.
- Following processing and extraction, the samples were subjected to 16S and 18s rRNA seq. Demultiplexed fastq files underwent quality trimming and denoising using the DADA2 pipeline.
- Alpha and beta diversity metrics were calculated using the phyloseq package (version –1.44).
- Network interactions were assessed using NetCoMi (version-1.1) and all graphical plots were constructed using ggplot2 (version-3.4.3).
- All downstream analyses were conducted in R software (version 4.2).

Results

- In a total of 160 samples, with 2 sets of 80 sites each undergoing sequencing, reads were rarefied to the minimum across all samples, resulting in 29,034(16s) and 11099(18s) Amplicon Sequence Variants (ASVs).
- The abundant phyla observed in water samples from both time points included prokaryotes like Proteobacteria, Planctomycetota and Bacteroidota and eukaryotes such as Gyrista and Ciliophora.
- Alpha diversity analysis revealed a significant difference for both prokaryotes and eukaryotes in water stored for varying durations regardless of the water source.

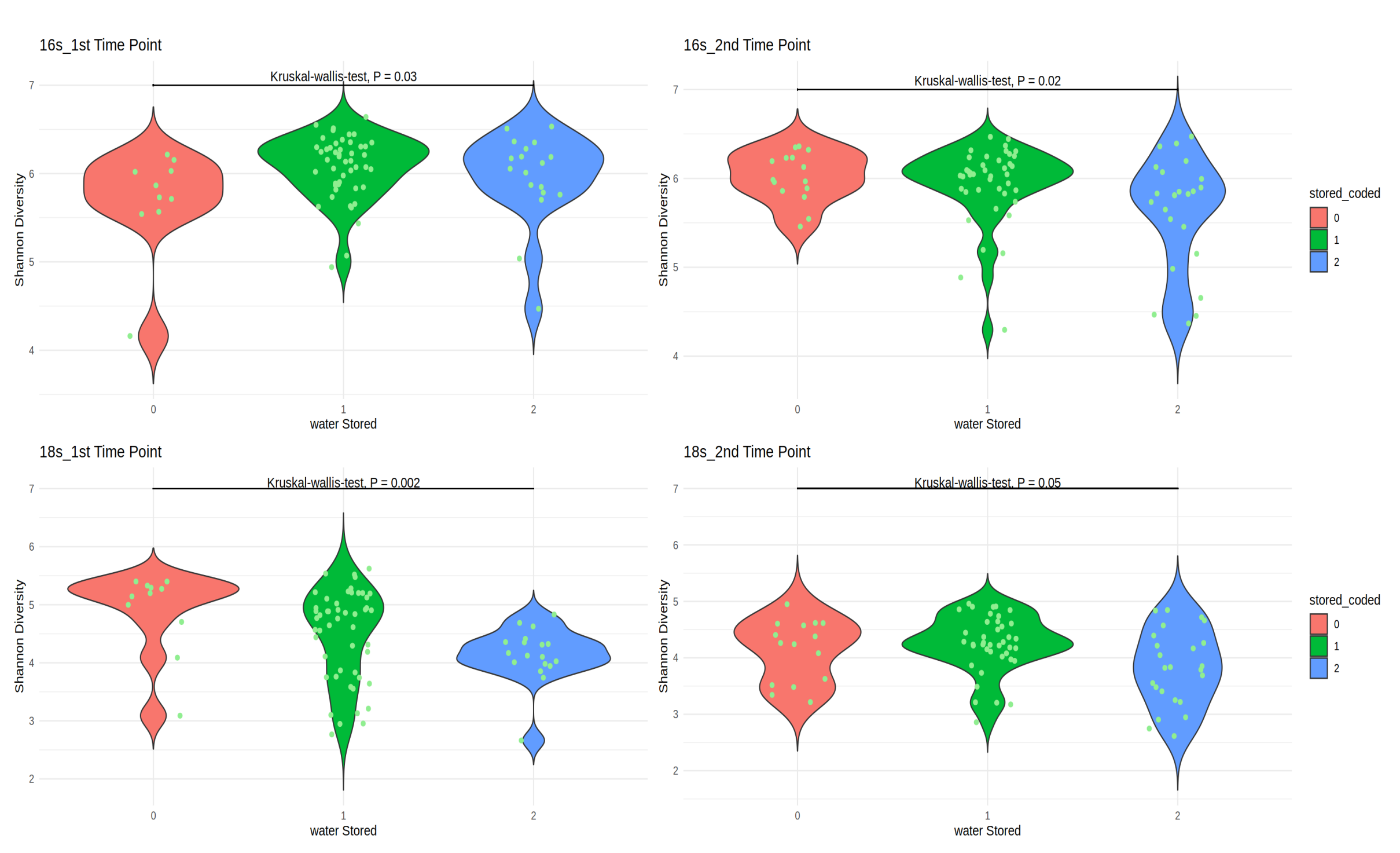


Figure 1: Shannon diversity (0- 0days, 1- <1 week, 2- >1 week)

- Environmental distance–decay relationship analysis revealed higher microbial diversity in specific communities and reduced diversity in distant sites, highlighting spatial variations in microbial community composition.
- Clustering analysis demonstrates diverse clusters even within the same localities.

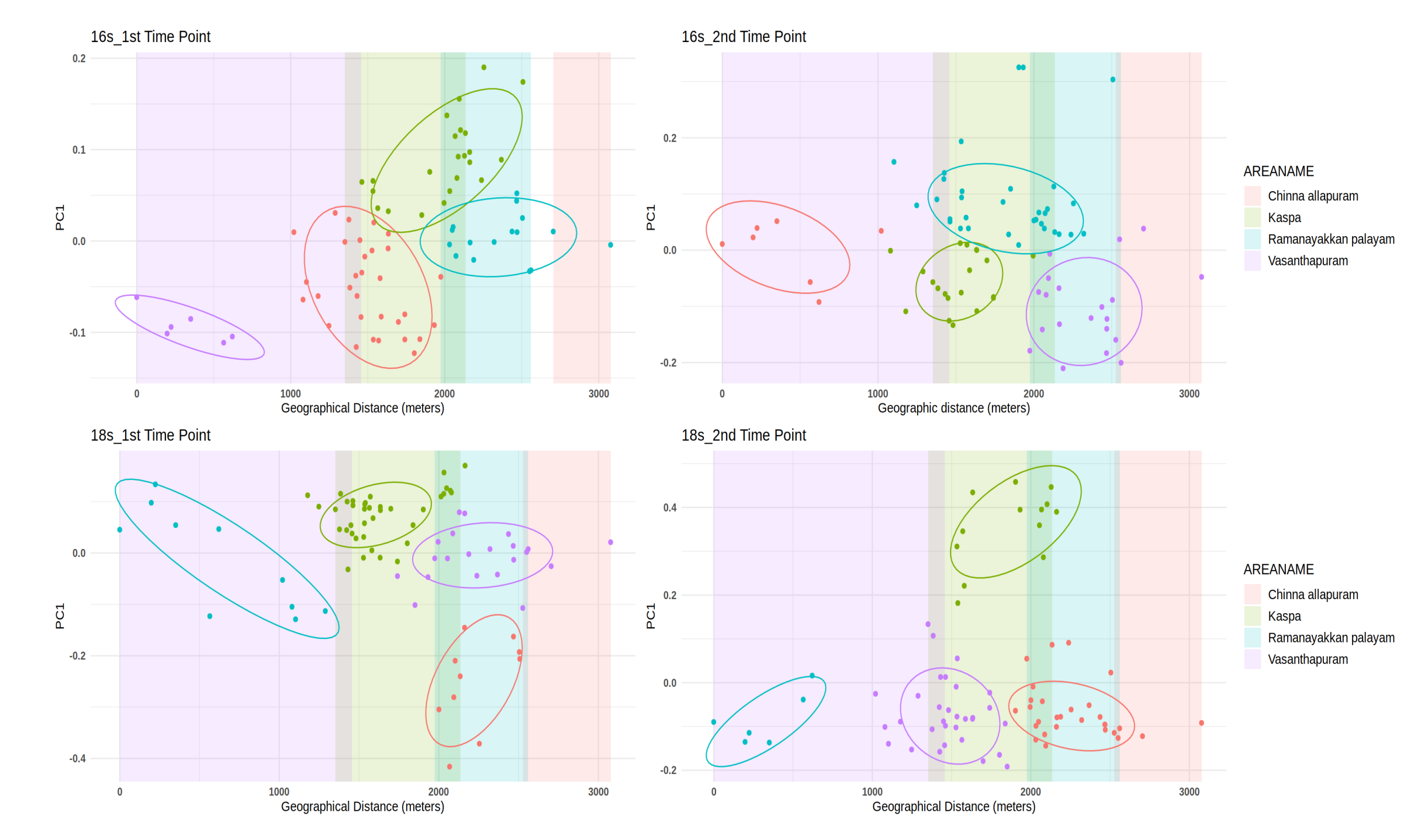


Figure 2: PC1 of Weighted Unifrac Distance against Geographical distance

- Network analysis results indicate temporal shifts in the topology structure with dynamic changes in degree centralities of both prokaryotes and eukaryotes.

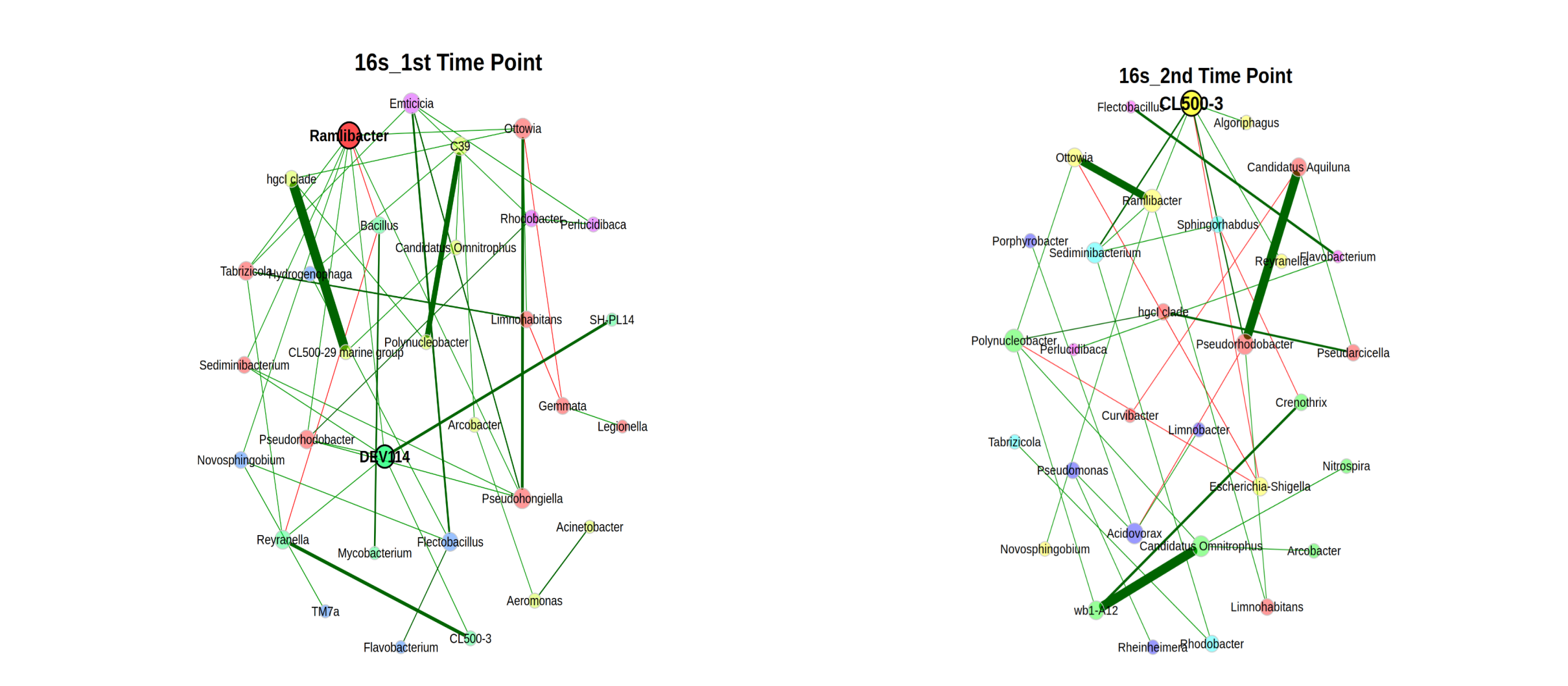


Figure 3: Network Topology of Prokaryotes (Green- Positive connections, Red- Negative connections)

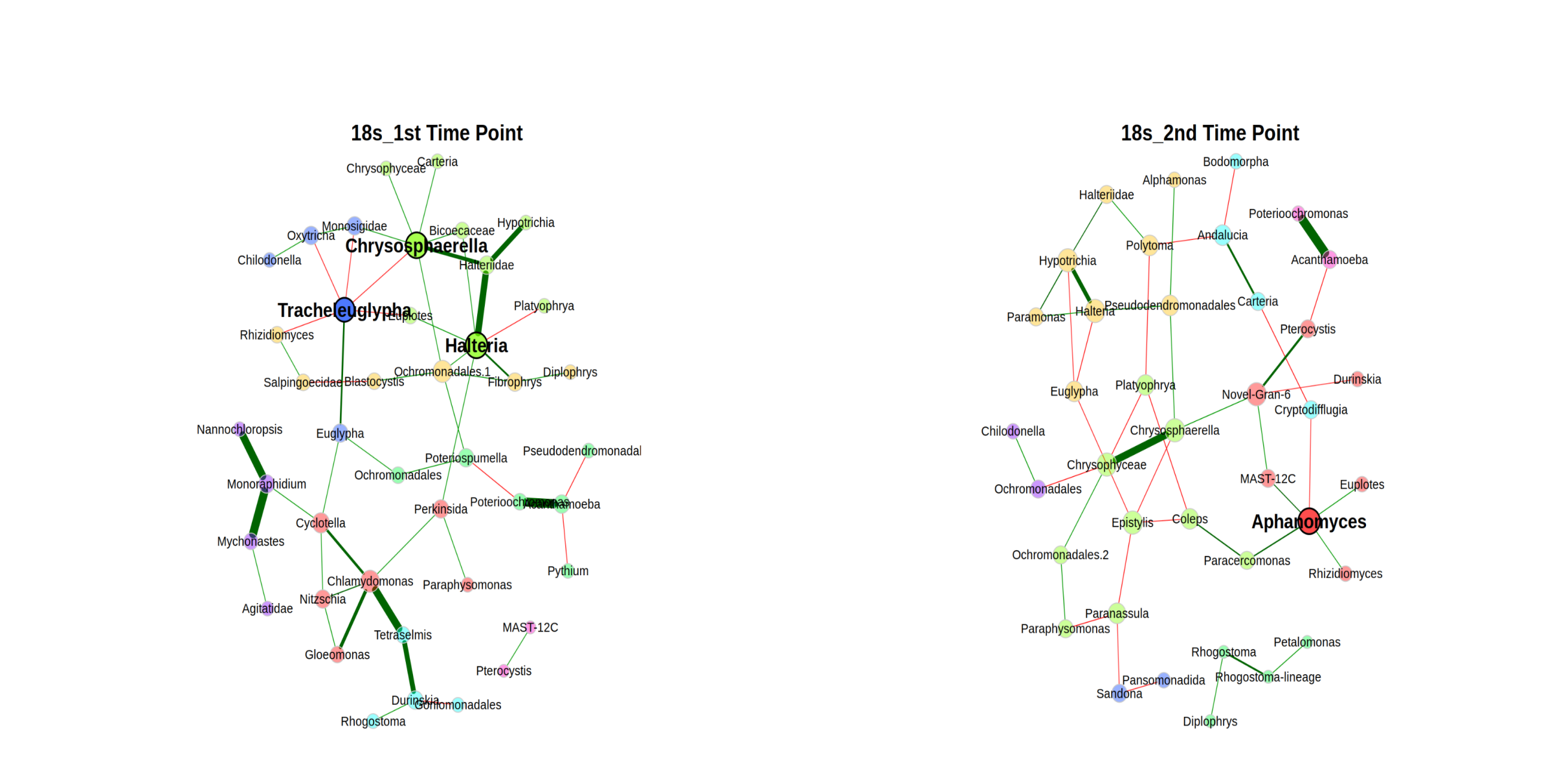


Figure 4: Network Topology of Eukaryotes (Green- Positive connections, Red- Negative connections)

- The cross-kingdom analysis showed lesser interspecies interactions between prokaryotic and eukaryotic communities.

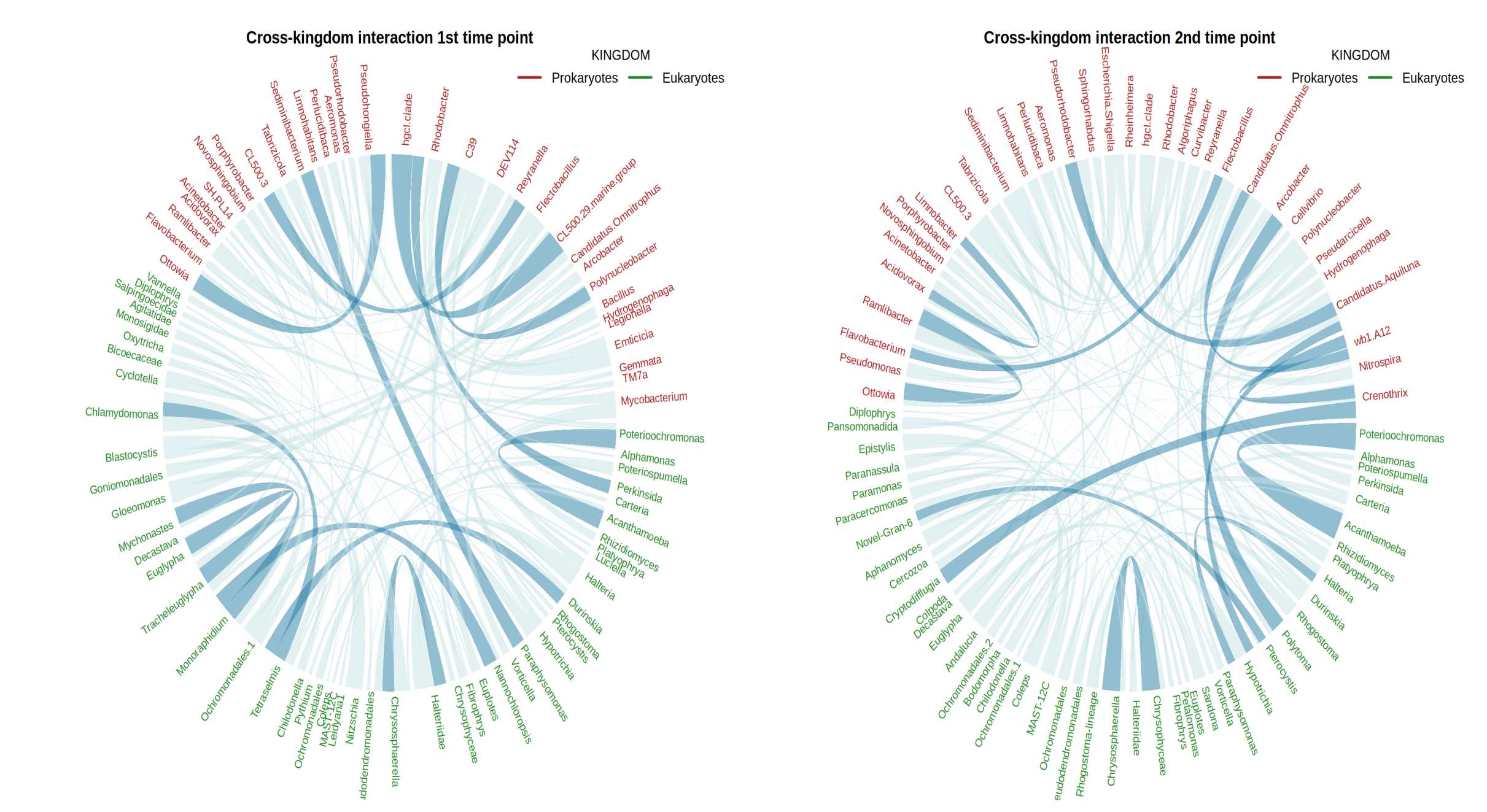


Figure 5: Crosskingdom Network Interaction of Prokaryotes and Eukaryotes (Darkblue - Strong association, Paleblue - Weak association)

Conclusion

- The microbial community structure differed over both space and time.
- Further sampling for a longer duration of time are essential to understand the community structure.
- Recognizing the spatial patterns of microbial diversity can guide the development of precise interventions to improve water quality in specific areas.
- These investigations are crucial for implementing effective strategies to safeguard public health and water resources.

Acknowledgement

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